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Task 01:

Implement Endpoint Security & Monitoring

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Implement Endpoint Security & Monitoring

Introduction:

In this task, the objective was to implement an Endpoint Detection and Response (EDR) solution to enhance system security and monitor activities across endpoints. A virtual lab environment was created consisting of a Kali Linux machine acting as the Wazuh server and a Windows 10 virtual machine acting as the monitored endpoint.

The purpose of this setup was to simulate a real-world Security Operations Center (SOC) environment, where endpoint activities are continuously monitored, analyzed, and logged. Tools such as Wazuh and Sysmon were used to detect suspicious activities, ensure file integrity, and generate alerts for potential threats. This task provided practical exposure to endpoint security, log analysis, and security monitoring techniques.

Objective:

The main objective of this task is to protect systems and endpoints from malware, unauthorized access, and suspicious activities by implementing an effective monitoring solution.

Key objectives include:

- Deploying an Endpoint Detection and Response (EDR) solution using Wazuh
- Monitoring system activity, file integrity, and user behavior
- Collecting and analyzing logs from endpoints in real-time
- Automating alerts for suspicious or malicious activities
- Utilizing Sysmon logs and integrating threat intelligence sources such as MalwareBazaar

Lab Setup:

A virtual lab environment was created to simulate a real-world monitoring system. The setup included:

- **Kali Linux Virtual Machine** acting as the Wazuh server
- **Windows 10 Virtual Machine** acting as the endpoint (agent)
- Both machines connected within the same virtual network for communication

The Kali machine was used to install and run the Wazuh manager and dashboard, while the Windows 10 system was configured with Sysmon and the Wazuh agent to generate and forward logs.

Tools & Technologies used:

Tools	Purpose
Wazuh	Acts as an EDR and SIEM platform used to monitor, analyze, and generate alerts from endpoint logs
Sysmon (Sysinternals)	Collects detailed Windows system activity such as process creation, network connections, and file changes
Kali Linux	Used as the central Wazuh server for monitoring and analysis
Windows 10 VM	Acts as the monitored endpoint where logs are generated
VMware	Used to create and manage virtual machines for the lab environment
MalwareBazaar	Used as a threat intelligence source for Indicators of Compromise (IOCs)

Security Monitoring Setup:

1. Virtual Environment Preparation

The first step involved setting up the virtual lab environment. Two virtual machines were created using VMware:

- Kali Linux VM (for Wazuh server)
- Windows 10 VM (for endpoint monitoring)

Both systems were configured on the same network to ensure communication between the agent and server.

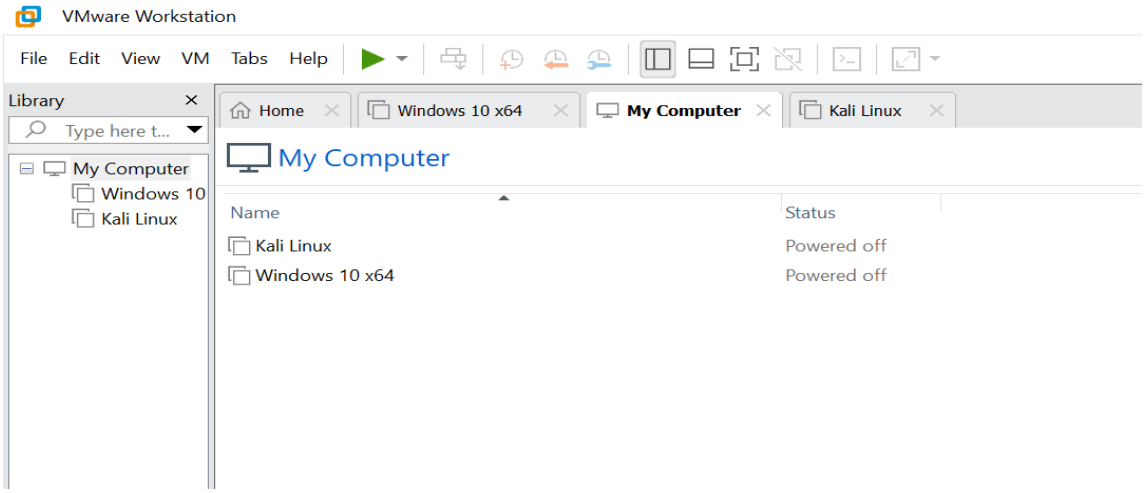

Figure: 01

2. Wazuh Installation on Kali Linux

Wazuh was installed on the Kali Linux system using the official installation script:

```bash
curl -sO https://packages.wazuh.com/4.7/wazuh-install.sh
sudo bash wazuh-install.sh -a -i
```

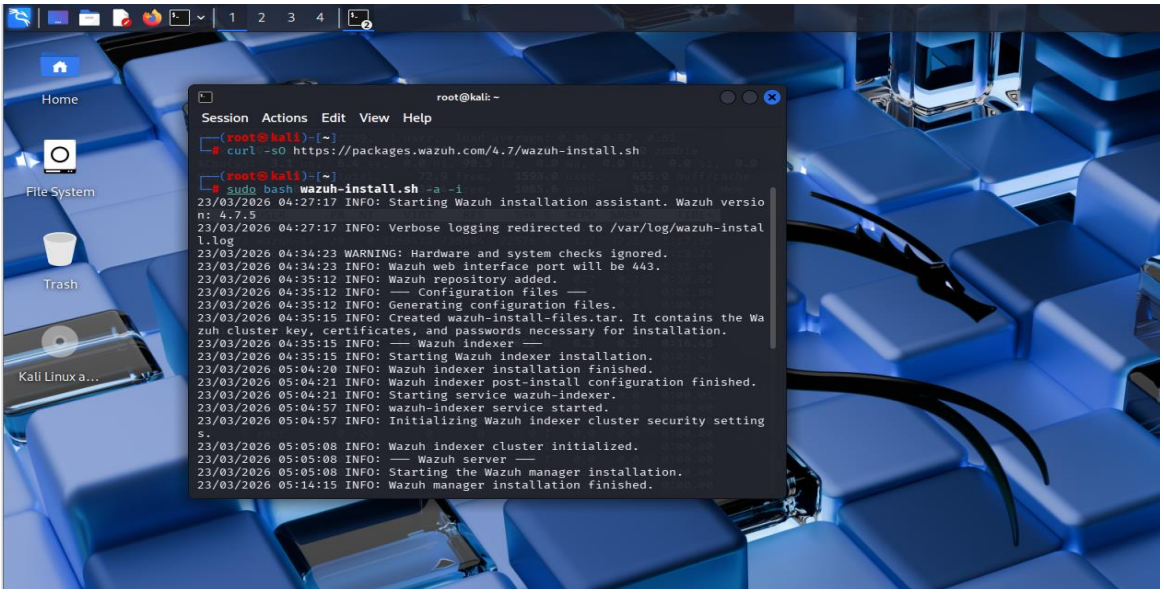

Figure: 02

The installation included:

- Wazuh Manager
- Wazuh Dashboard
- Wazuh Indexer

After installation, the Wazuh dashboard was accessed via browser using:

```
https://<Kali-IP>
```

A security warning may appear in the browser due to a self-signed certificate. This can be safely ignored by proceeding to the dashboard.

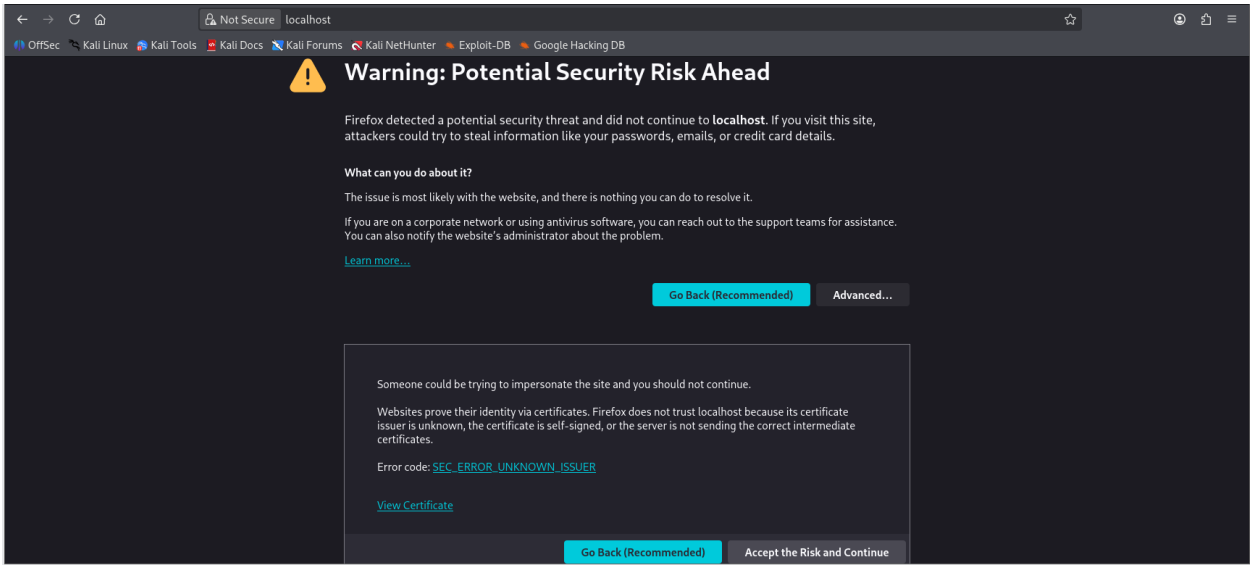

Figure: 03

Wazuh server will be opened demanding the username and password generated upon completion of wazuh server installation, Enter the username and password.

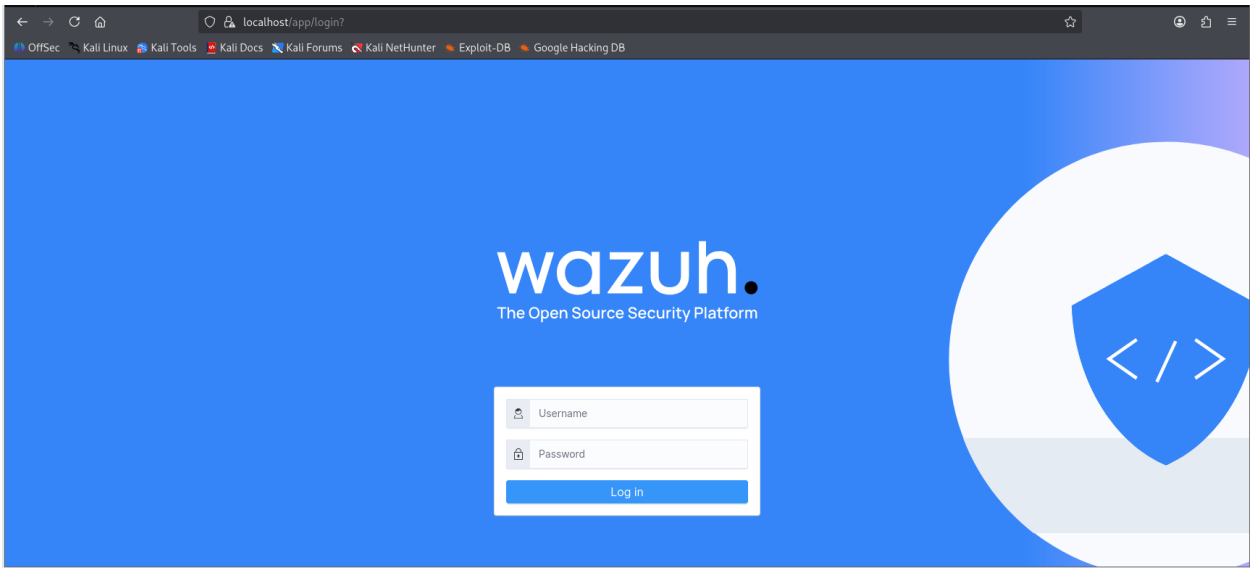

Figure: 04

3. Sysmon Installation on Windows 10

Sysmon was downloaded from Sysinternals and installed along with a configuration file.

Steps performed:

1. Download Sysmon and extract files
2. Download Sysmon configuration file (sysmonconfig-export.xml)
3. Place configuration file in the same folder

Installation command (run in Command Prompt as Administrator):

```cmd
sysmon.exe -accepteula -i sysmonconfig-export.xml
```

Sysmon was successfully installed and started generating detailed system logs.

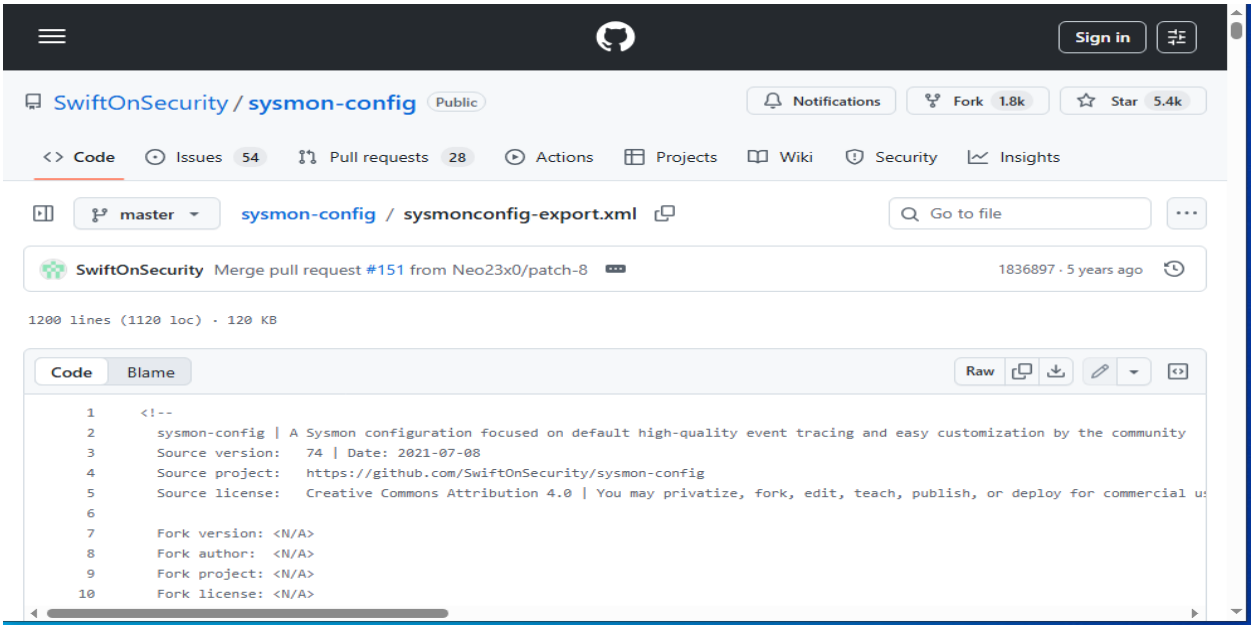

Figure: 05

4. Wazuh Agent Installation on Windows 10

The Wazuh agent was installed on the Windows VM using the MSI installer.

Installation command (PowerShell as Administrator):

```PowerShell
.\wazuh-agent-4.x.x.msi /q WAZUH_MANAGER="192.168.x.x" WAZUH_AGENT_NAME="Wi
```

.\wazuh-agent-4.x.x.msi /q WAZUH_MANAGER="192.168.x.x" WAZUH_AGENT_NAME="Windows_Agent1"

After installation, the agent service was started using:

```cmd
NET START WazuhSvc
```

This enabled secure communication between the Windows endpoint and the Wazuh server.

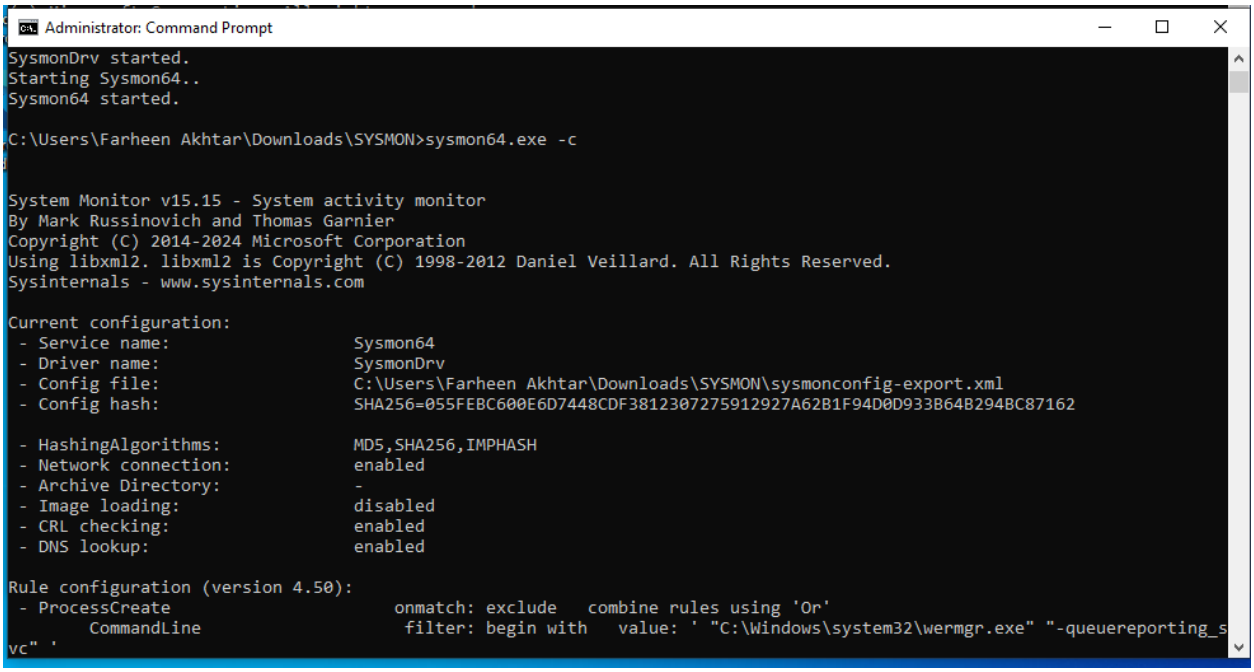

Figure: 06

5. Adding Agent to Wazuh Dashboard

The agent was configured to connect with the Wazuh server using:

- Server IP (Kali machine IP)
- Agent name
- Default group

Once configured and service was running, the agent attempted to register with the Wazuh manager.

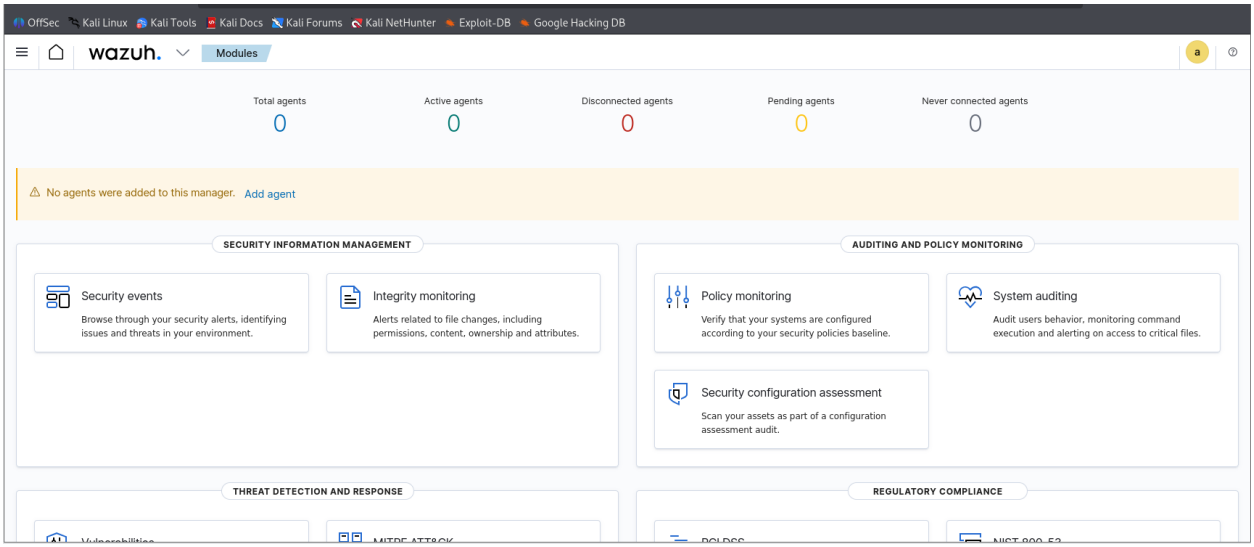

Figure: 07

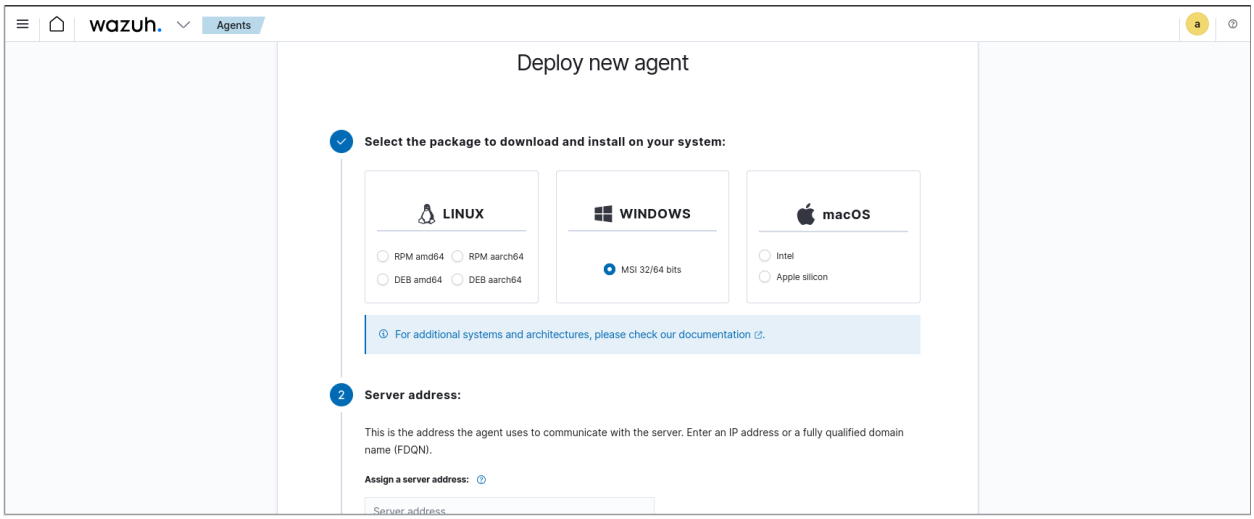

Figure: 08

6. Monitoring and Alerts

After successful setup:

- Sysmon generated detailed logs
- Wazuh agent forwarded logs to the server
- Wazuh analyzed logs and generated alerts for suspicious activities

The dashboard provided visibility into:

- System events
- Security alerts
- Endpoint activity

This setup simulates a basic Security Operations Center (SOC) workflow, where logs are collected, analyzed, and alerts are generated for potential threats.

Challenges faced:

During the implementation of this task, several challenges were encountered:

- Internet connectivity issues during Wazuh installation
- Wazuh installation compatibility issues on Kali Linux
- Difficulty in accessing the Wazuh dashboard initially
- Problems with Wazuh agent installation and service creation on Windows
- Agent not appearing on dashboard due to configuration issues
- Confusion between Sysmon and Wazuh agent roles
- Firewall and connectivity issues between virtual machines

These challenges helped in understanding real-world troubleshooting in security environments.

Learning outcomes:

This task provided valuable learning experience in the field of cybersecurity and SOC operations:

- Understanding of Endpoint Detection and Response (EDR) systems
- Hands-on experience with Wazuh as a SIEM tool
- Knowledge of Sysmon for detailed Windows logging
- Understanding of log collection, monitoring, and alerting
- Importance of proper configuration in security tools
- Practical exposure to troubleshooting and system integration
- Awareness of real-time monitoring and threat detection techniques

Conclusion:

In conclusion, this task successfully demonstrated the implementation of endpoint security and monitoring using Wazuh and Sysmon. A virtual lab environment was created to simulate real-world scenarios where endpoint activities are monitored and analyzed.

The integration of Sysmon with Wazuh enabled detailed log collection and analysis, providing visibility into system behavior and potential threats. Despite encountering several challenges, the task helped in building a strong foundation in security monitoring, log analysis, and EDR deployment.

This practical experience is highly valuable for understanding modern cybersecurity practices and SOC operations.